

■ 시나리오 별 백업 복구(rman using recovery catalog)

KEY WORDS

- <시나리오 17> Recovering Tables
- <시나리오 18> Rman 암호화
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- <시나리오 20> 파라미터 파일(spfile, pfile)이 손상되었을 경우 복구
- <시나리오 21> Incremental Backup

<<시나리오 17>> Recovering Tables

- 특정한 테이블스페이스만 이용해서 복제 데이터베이스 생성 한 후 복구

```
SYS@orcl> create tablespace insa_tbs datafile '/u01/app/oracle/oradata/ORCL/insa_tbs01' size 10m;
Tablespace created.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	4938227	SYSTEM
2	OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf	4938227	ONLINE
8	OETBS	/u01/app/oracle/oradata/ORCL/oetbs02.dbf	4938227	ONLINE
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	4938227	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	4938227	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	4938227	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	4938227	ONLINE
9	INSA_TBS	/u01/app/oracle/oradata/ORCL/insa_tbs01	4948259	ONLINE

8 rows selected.

```
SYS@orcl> create table hr.insa_2025 tablespace insa_tbs as select * from hr.employees;
Table created.
```

```
SYS@orcl> select count(*) from hr.insa_2025;
COUNT(*)
```

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```
RMAN> report schema;
```

starting full resync of recovery catalog

full resync complete

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	630	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf
9	10	INSA_TBS	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32	/u01/app/oracle/oradata/ORCL/temp01.dbf

RMAN> backup as compressed backupset database;

Starting backup at 20-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=1 device type=DISK

channel ORA_DISK_1: starting compressed full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf

input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf

input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf

input datafile file number=00002 name=/u01/app/oracle/oradata/ORCL/oetbs01.dbf

input datafile file number=00005 name=/u01/app/oracle/oradata/ORCL/shtbs01.dbf

input datafile file number=00008 name=/u01/app/oracle/oradata/ORCL/oetbs02.dbf

input datafile file number=00009 name=/u01/app/oracle/oradata/ORCL/insa_tbs01

input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: starting piece 1 at 20-MAY-25

channel ORA_DISK_1: finished piece 1 at 20-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bk

p tag=TAG20250520T225036 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:35
 Finished backup at 20-MAY-25

Starting Control File and SPFILE Autobackup at 20-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201647072_n2s26jvd_.bkp
 comment=NONE

Finished Control File and SPFILE Autobackup at 20-MAY-25

RMAN> report obsolete;

RMAN> delete obsolete;

RMAN> crosscheck archivelog all;

released channel: ORA_DISK_1

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=1 device type=DISK

specification does not match any archived log in the repository

RMAN> list expired archivelog all;

specification does not match any archived log in the repository

RMAN> delete expired archivelog all;

released channel: ORA_DISK_1

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=1 device type=DISK

specification does not match any archived log in the repository

RMAN> alter system archive log current;

Statement processed

RMAN> list archivelog all;

List of Archived Log Copies for database with db_unique_name ORCL

=====

Key	Thrd	Seq	S	Low Time
-----	------	-----	---	----------

3285	1	10	A	20-MAY-25
------	---	----	---	-----------

Name: /home/oracle/arch1/arch_1_10_1201567818.arc

RMAN> list backup;

List of Backup Sets

=====

BS Key	Type	LV	Size	Device Type	Elapsed Time	Completion Time
--------	------	----	------	-------------	--------------	-----------------

3240 Full 305.41M DISK 00:00:30 20-MAY-25

BP Key: 3243 Status: AVAILABLE Compressed: YES Tag: TAG20250520T225036

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp

List of Datafiles in backup set 3240

File LV Type Ckp SCN Ckp Time Abs Fuz SCN Sparse Name

1	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/system01.dbf
2	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf
9	Full	4950913	20-MAY-25	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01

BS Key Type LV Size Device Type Elapsed Time Completion Time

3257 Full 10.52M DISK 00:00:00 20-MAY-25

BP Key: 3266 Status: AVAILABLE Compressed: NO Tag: TAG20250520T225112

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201647072_n2s26jvd_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 4951003 Ckp time: 20-MAY-25

SYS@orcl> select systimestamp from dual;

SYSTIMESTAMP

20-MAY-25 10.59.59.927236 PM +09:00

SYS@orcl> drop table hr.insa_2025 purge; <--- 아~~ 이런 엄청난 실수를 ㅠㅠ.

Table dropped.

SYS@orcl> alter system archive log current;

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP# SEQUENCE#

MEMBER

```

-----
      MB STATUS      ARC FIRST_CHANGE# NEXT_CHANGE#
-----
      1          10
/u01/app/oracle/oradata/ORCL/redo01.log
      10 INACTIVE      YES      4947662      4951817

      2          11
/u01/app/oracle/oradata/ORCL/redo02.log
      10 ACTIVE        YES      4951817      4952696

```

GROUP# SEQUENCE#

MEMBER

```

-----
      MB STATUS      ARC FIRST_CHANGE# NEXT_CHANGE#
-----
      3          12
/u01/app/oracle/oradata/ORCL/redo03.log
      10 CURRENT      NO      4952696      1.8447E+19

```

SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v\$archived_log;

```

SEQUENCE#      NAME                                     FIRST_CHANGE# NEXT_CHANGE#
-----
      10      /home/oracle/arch1/arch_1_10_1201567818.arc  4947662      4951817
      11      /home/oracle/arch1/arch_1_11_1201567818.arc  4951817      4952696

```

SYS@orcl> ! ls /home/oracle/arch1/
arch_1_10_1201567818.arc arch_1_11_1201567818.arc

백업 데이터파일 복사

[oracle@oel7v9r1 2025_05_20]\$ pwd

/home/oracle/backup/FRA/ORCL/backupset/2025_05_20

[oracle@oel7v9r1 2025_05_20]\$ cp -v o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp /home/oracle/ora19c/

'o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp'

->

'/home/oracle/ora19c/o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp'

* 백업 컨트롤파일 복사

```
[oracle@oel7v9r1 2025_05_20]$ pwd
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20
[oracle@oel7v9r1 2025_05_20]$ ls -al
total 10784
drwxr-x---. 2 oracle dba      46 May 20 22:55 .
drwxr-x---. 6 oracle dba      78 May 20 00:05 ..
-rw-r-----. 1 oracle dba 11042816 May 20 22:51 o1_mf_s_1201647072_n2s26jvd_.bkp
[oracle@oel7v9r1 2025_05_20]$ cp -v o1_mf_s_1201647072_n2s26jvd_.bkp /home/oracle/ora19c/
'o1_mf_s_1201647072_n2s26jvd_.bkp' -> '/home/oracle/ora19c/o1_mf_s_1201647072_n2s26jvd_.bkp'
```

아카이브 파일 복사

```
[oracle@oel7v9r1 ~]$ cp -v /home/oracle/arch1/* /home/oracle/ora19c/
'/home/oracle/arch1/arch_1_10_1201567818.arc' -> '/home/oracle/ora19c/arch_1_10_1201567818.arc'
'/home/oracle/arch1/arch_1_11_1201567818.arc' -> '/home/oracle/ora19c/arch_1_11_1201567818.arc'
```

- 복제 DB(SID: ora19c)에 접속

```
[oracle@oel7v9r1 ~]$ . oraenv
ORACLE_SID = [orcl] ? ora19c
ORACLE_HOME = [/home/oracle] ? /u01/app/oracle/product/19.3.0/dbhome_1
The Oracle base remains unchanged with value /u01/app/oracle
[oracle@oel7v9r1 ~]$ sqlplus / as sysdba
SYS@ora19c> startup pfile=/home/oracle/ora19c/initora19c.ora nomount
ORACLE instance started.
Total System Global Area  243268216 bytes
Fixed Size                  8895096 bytes
Variable Size              180355072 bytes
Database Buffers           50331648 bytes
Redo Buffers                3686400 bytes
SYS@ora19c>
```

rman 데이터베이스 파일 복원

필수 테이블 스페이스 : system, sysaux, undotbs

```
RMAN> run{
2 duplicate target database to 'ora19c'
3 tablespace insa_tbs
4 pfile='/home/oracle/ora19c/initora19c.ora'
5 nofilenamecheck
6 backup location '/home/oracle/ora19c'
7until time "to_date('2025-05-20 22:59:50','yyyy-mm-dd hh24:mi:ss')";
```

8 }

<<시나리오 >> Rman 암호화

RMAN> list backup;

using target database control file instead of recovery catalog

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

35	Full		305.41M	DISK		00:00:30	20-MAY-25
----	------	--	---------	------	--	----------	-----------

BP Key: 35 Status: AVAILABLE Compressed: YES Tag: TAG20250520T225036

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp

List of Datafiles in backup set 35

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

1	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/system01.dbf
2	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/users01.dbf
8	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/oetbs02.dbf
9	Full	4950913	20-MAY-25		NO		/u01/app/oracle/oradata/ORCL/insa_tbs01

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

36	Full		10.52M	DISK		00:00:00	20-MAY-25
----	------	--	--------	------	--	----------	-----------

BP Key: 36 Status: AVAILABLE Compressed: NO Tag: TAG20250520T225112

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201647072_n2s26jvd_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 4951003 Ckp time: 20-MAY-25

RMAN> report need backup;

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of files with less than 1 redundant backups

File #bkps Name

RMAN> show encryption algorithm;

RMAN configuration parameters for database with db_unique_name ORCL are:

CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default

RMAN> select algorithm_id, algorithm_name, algorithm_description, is_default
from v\$rman_encryption_algorithms;

2>

ALGORITHM_ID ALGORITHM_NAME

ALGORITHM_DESCRIPTION IS_

1 AES128	
AES 128-bit key	YES
2 AES192	
AES 192-bit key	NO
3 AES256	
AES 256-bit key	NO

RMAN> set encryption on identified by oracle only;

executing command: SET encryption

RMAN> list backup;

RMAN> select handle, tag, encrypted from v\$backup_piece;

HANDLE	TAG	ENC

/home/oracle/backup/FRA/ORCL/backupset/2025_05_20/o1_mf_nnndf_TAG20250520T225036_n2s25f0s_.bkp		
TAG20250520T225036	NO	
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_20/o1_mf_s_1201647072_n2s26jvd_.bkp		
TAG20250520T225112	NO	
/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T000525_n2s6kp7x_.bkp		
TAG20250521T000525	YES	
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201651561_n2s6lsc1_.bkp		
TAG20250521T000601	YES	

38 rows selected

암호화 백업본 사용

SYS@orcl> select name from v\$datafile;

NAME

/u01/app/oracle/oradata/ORCL/system01.dbf

/u01/app/oracle/oradata/ORCL/oetbs01.dbf

/u01/app/oracle/oradata/ORCL/sysaux01.dbf

/u01/app/oracle/oradata/ORCL/undotbs01.dbf

/u01/app/oracle/oradata/ORCL/shtbs01.dbf

/u01/app/oracle/oradata/ORCL/users01.dbf

/u01/app/oracle/oradata/ORCL/oetbs02.dbf

/u01/app/oracle/oradata/ORCL/insa_tbs01

8 rows selected.

SYS@orcl> shutdown immediate

Database closed.

Database dismounted.

ORACLE instance shut down.

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf

RMAN> list failure;

using target database control file instead of recovery catalog

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	21-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

RMAN> list failure 2 detail;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	21-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

7965	HIGH	OPEN	21-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is
------	------	------	-----------	---

missing

Impact: Some objects in tablespace USERS might be unavailable

RMAN> restore datafile 7;

Starting restore at 21-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=73 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel	ORA_DISK_1:	reading	from	backup	piece
---------	-------------	---------	------	--------	-------

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T000525_n2s6kp7x_.bkp

RMAN-00571: =====

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====

RMAN-00571: =====

RMAN-03002: failure of restore command at 05/21/2025 00:15:09

ORA-19870:	error	while	restoring	backup	piece
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/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T000525_n2s6kp7x_.bkp

ORA-19913: unable to decrypt backup

ORA-28365: wallet is not open <-- --> 백업본이 암호화되어 있기 때문에 restore시에 에러 발생.

RMAN> set decryption identified by oracle; <--- 암호 설정

executing command: SET decryption

RMAN> restore datafile 7;

Starting restore at 21-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel	ORA_DISK_1:	reading	from	backup	piece
---------	-------------	---------	------	--------	-------

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T000525_n2s6kp7x_.bkp

channel	ORA_DISK_1:				piece
---------	-------------	--	--	--	-------

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T000525_n2s6kp7x_.b

kp tag=TAG20250521T000525

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 21-MAY-25

RMAN> recover datafile 7;

Starting recover at 21-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:00

Finished recover at 21-MAY-25

RMAN> alter database datafile 7 online;

Statement processed

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	630	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf
9	10	INSA_TBS	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32	/u01/app/oracle/oradata/ORCL/temp01.dbf

<<시나리오 19>> Image copy

RMAN> backup as copy database;

Starting backup at 21-MAY-25
using channel ORA_DISK_1
skipping datafile 5; already backed up 2 time(s)
channel ORA_DISK_1: starting datafile copy
input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf
output file name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_system_n2ty43ty_.dbf
tag=TAG20250521T155355 RECID=7 STAMP=1201708468
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:35
channel ORA_DISK_1: starting datafile copy
input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf
output file name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_sysaux_n2ty56j0_.dbf
tag=TAG20250521T155355 RECID=8 STAMP=1201708495
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:25
channel ORA_DISK_1: starting datafile copy
input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf
output file name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_undotbs1_n2ty5zxv_.dbf
tag=TAG20250521T155355 RECID=9 STAMP=1201708516
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:27
channel ORA_DISK_1: starting datafile copy
input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf
output file name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2ty6tow_.dbf tag=TAG20250521T155355
RECID=10 STAMP=1201708528
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:07
Finished backup at 21-MAY-25

Starting Control File and SPFILE Autobackup at 21-MAY-25
piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201708532_n2ty743s_.bkp
comment=NONE
Finished Control File and SPFILE Autobackup at 21-MAY-25

RMAN> backup as copy current controlfile;

Starting backup at 21-MAY-25
using channel ORA_DISK_1
channel ORA_DISK_1: starting datafile copy
copying current control file

output file name=/home/oracle/backup/FRA/ORCL/controlfile/o1_mf_TAG20250521T155541_n2ty7f9y_.ctl
tag=TAG20250521T155541 RECID=11 STAMP=1201708541
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:01
Finished backup at 21-MAY-25

Starting Control File and SPFILE Autobackup at 21-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201708542_n2ty7gwr_.bkp
comment=NONE

Finished Control File and SPFILE Autobackup at 21-MAY-25

RMAN> list copy;

List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
3636	1 A	21-MAY-25	3206578	21-MAY-25	NO
	Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_system_n2ty43ty_.dbf				
	Tag: TAG20250521T155355				
3637	3 A	21-MAY-25	3206594	21-MAY-25	NO
	Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_sysaux_n2ty56j0_.dbf				
	Tag: TAG20250521T155355				
3638	4 A	21-MAY-25	3206603	21-MAY-25	NO
	Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_undotbs1_n2ty5zxv_.dbf				
	Tag: TAG20250521T155355				
3639	7 A	21-MAY-25	3206618	21-MAY-25	NO
	Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2ty6tow_.dbf				
	Tag: TAG20250521T155355				

List of Control File Copies

=====

Key	S	Completion Time	Ckp SCN	Ckp Time
3728	A	21-MAY-25	3206676	21-MAY-25
	Name: /home/oracle/backup/FRA/ORCL/controlfile/o1_mf_TAG20250521T155541_n2ty7f9y_.ctl			

Tag: TAG20250521T155541

List of Archived Log Copies for database with db_unique_name ORCL

=====

Key	Thrd	Seq	S	Low	Time

2769	1	101	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000101_1200585133.arc					
2768	1	101	A	21-MAY-25	
Name: /home/oracle/arch1/0001_0000000101_1200585133.arc					
2995	1	102	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000102_1200585133.arc					
2994	1	102	A	21-MAY-25	
Name: /home/oracle/arch1/0001_0000000102_1200585133.arc					
3167	1	103	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000103_1200585133.arc					
3166	1	103	A	21-MAY-25	
Name: /home/oracle/arch1/0001_0000000103_1200585133.arc					
3288	1	104	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000104_1200585133.arc					
3287	1	104	A	21-MAY-25	
Name: /home/oracle/arch1/0001_0000000104_1200585133.arc					
3318	1	105	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000105_1200585133.arc					
3317	1	105	A	21-MAY-25	
Name: /home/oracle/arch1/0001_0000000105_1200585133.arc					
3416	1	106	A	21-MAY-25	
Name: /home/oracle/arch2/0001_0000000106_1200585133.arc					

3415 1 106 A 21-MAY-25

Name: /home/oracle/arch1/0001_0000000106_1200585133.arc

3498 1 107 A 21-MAY-25

Name: /home/oracle/arch2/0001_0000000107_1200585133.arc

3497 1 107 A 21-MAY-25

Name: /home/oracle/arch1/0001_0000000107_1200585133.arc

RMAN> list copy of tablespace system;

List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
3636	1	A 21-MAY-25	3206578	21-MAY-25	NO
Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_system_n2ty43ty_.dbf					
Tag: TAG20250521T155355					

RMAN> list copy of controlfile;

List of Control File Copies

=====

Key	S	Completion Time	Ckp SCN	Ckp Time
3728	A	21-MAY-25	3206676	21-MAY-25
Name: /home/oracle/backup/FRA/ORCL/controlfile/o1_mf_TAG20250521T155541_n2ty7f9y_.ctl				
Tag: TAG20250521T155541				

SQL> shut immediate

- 장애 발생

SQL> !rm /u01/app/oracle/oradata/ORCL/users01.dbf

SQL> startup

ORACLE instance started.

Total System Global Area 1811936368 bytes
 Fixed Size 8897648 bytes
 Variable Size 436207616 bytes
 Database Buffers 1358954496 bytes
 Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 7 - see DBWR trace file

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

RMAN> list failure;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	21-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

RMAN> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME	CON_ID
7	ONLINE	ONLINE	FILE NOT FOUND	0	0

RMAN> alter database datafile 7 offline;

Statement processed

RMAN> alter database open;

Statement processed

RMAN> list failure 2 detail;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID Priority Status Time Detected Summary

Failure ID	Priority	Status	Time Detected	Summary
2	HIGH	OPEN	21-MAY-25	One or more non-system datafiles are missing

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID Priority Status Time Detected Summary

Failure ID	Priority	Status	Time Detected	Summary
2165	HIGH	OPEN	21-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is missing

Impact: Some objects in tablespace USERS might be unavailable

RMAN> list copy of datafile 7

2> ;

List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
3639	7 A	21-MAY-25	3206618	21-MAY-25	NO

Name: /home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2ty6tow_.dbf
Tag: TAG20250521T155355

RMAN> list backup;

List of Backup Sets

=====

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
2908	Incr 0	144.03M	DISK		00:00:11	21-MAY-25

BP Key: 2913 Status: AVAILABLE Compressed: NO Tag: TAG20250521T100158
Piece Name:
/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnnd0_TAG20250521T100158_n2t9j7fw_.bkp

List of Datafiles in backup set 2908

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

7	0	Incr	3192299	21-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf
---	---	------	---------	-----------	--	----	--

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time

2909	Incr 0	530.29M	DISK	00:00:32	21-MAY-25
BP Key: 2914 Status: AVAILABLE Compressed: NO Tag: TAG20250521T100158					
Piece Name:					
/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnnd0_TAG20250521T100158_n2t9j7g6_.bkp					
List of Datafiles in backup set 2909					
File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN Sparse Name

3	0	Incr	3192298	21-MAY-25	NO /u01/app/oracle/oradata/ORCL/sysaux01.dbf
5	0	Incr	3166776	20-MAY-25	NO /u01/app/oracle/oradata/ORCL/ts01.dbf

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time

2954	Full	10.52M	DISK	00:00:01	21-MAY-25
BP Key: 2957 Status: AVAILABLE Compressed: NO Tag: TAG20250521T100301					
Piece Name:					
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201687381_n2t9l604_.bkp					
SPFILE Included: Modification time: 20-MAY-25					
SPFILE db_unique_name: ORCL					
Control File Included: Ckp SCN: 3192338 Ckp time: 21-MAY-25					

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time

3047	Full	530.38M	DISK	00:01:37	21-MAY-25
BP Key: 3050 Status: AVAILABLE Compressed: NO Tag: TAG20250521T112348					
Piece Name:					
/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T112348_n2tg9o9d_.bkp					
List of Datafiles in backup set 3047					
File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN Sparse Name

3		Full	3196009	21-MAY-25	NO /u01/app/oracle/oradata/ORCL/sysaux01.dbf

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time

3070	Full	10.52M	DISK	00:00:01	21-MAY-25
BP Key: 3072 Status: AVAILABLE Compressed: NO Tag: TAG20250521T112537					
Piece Name:					
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201692337_n2tgf1w2_.bkp					

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3196136 Ckp time: 21-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

3095	Full	1.60M	DISK		00:00:01	21-MAY-25
------	------	-------	------	--	----------	-----------

BP Key: 3098 Status: AVAILABLE Compressed: NO Tag: TAG20250521T112539

Piece Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T112539_n2tgf3g8_.bkp

List of Datafiles in backup set 3095

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

4	Full	3196168	21-MAY-25		NO		/u01/app/oracle/oradata/ORCL/undotbs01.dbf
---	------	---------	-----------	--	----	--	--

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

3120	Full	10.52M	DISK		00:00:01	21-MAY-25
------	------	--------	------	--	----------	-----------

BP Key: 3122 Status: AVAILABLE Compressed: NO Tag: TAG20250521T112542

Piece Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201692342_n2tgf70q_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3196182 Ckp time: 21-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

3223	Full	1.42G	DISK		00:04:38	21-MAY-25
------	------	-------	------	--	----------	-----------

List of Datafiles in backup set 3223

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

1	Full	3200237	21-MAY-25		NO		/u01/app/oracle/oradata/ORCL/system01.dbf
3	Full	3200237	21-MAY-25		NO		/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	Full	3200237	21-MAY-25		NO		/u01/app/oracle/oradata/ORCL/undotbs01.dbf
7	Full	3200237	21-MAY-25		NO		/u01/app/oracle/oradata/ORCL/users01.dbf

Backup Set Copy #1 of backup set 3223

Device	Type	Elapsed Time	Completion Time	Compressed	Tag
--------	------	--------------	-----------------	------------	-----

DISK		00:04:38	21-MAY-25	NO	TAG20250521T130734
------	--	----------	-----------	----	--------------------

List of Backup Pieces for backup set 3223 Copy #1

BP Key	Pc#	Status	Piece Name
3226	1	AVAILABLE	/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T130734_n2tnd6ww_.bkp
3227	2	AVAILABLE	/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T130734_n2tnlz6j_.bkp

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
3254	Full		10.52M	DISK		00:00:01	21-MAY-25

BP Key: 3259 Status: AVAILABLE Compressed: NO Tag: TAG20250521T131233

Piece Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201698753_n2tnol75_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3200374 Ckp time: 21-MAY-25

BS Key	Size	Device	Type	Elapsed Time	Completion Time
3451	29.31M	DISK		00:00:04	21-MAY-25

BP Key: 3454 Status: AVAILABLE Compressed: NO Tag: TAG20250521T155022

Piece Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_annnn_TAG20250521T155022_n2txh5r_.bkp

List of Archived Logs in backup set 3451

Thrd	Seq	Low SCN	Low Time	Next SCN	Next Time
1	101	3188218	21-MAY-25 3190772	21-MAY-25	
1	102	3190772	21-MAY-25 3192367	21-MAY-25	
1	103	3192367	21-MAY-25 3197274	21-MAY-25	
1	104	3197274	21-MAY-25 3201927	21-MAY-25	
1	105	3201927	21-MAY-25 3204441	21-MAY-25	
1	106	3204441	21-MAY-25 3206353	21-MAY-25	

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
3499	Full		1.42G	DISK		00:01:16	21-MAY-25

List of Datafiles in backup set 3499

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
1		Full	3206388	21-MAY-25		NO	/u01/app/oracle/oradata/ORCL/system01.dbf
3		Full	3206388	21-MAY-25		NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4		Full	3206388	21-MAY-25		NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
7		Full	3206388	21-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf

Backup Set Copy #1 of backup set 3499

Device	Type	Elapsed Time	Completion Time	Compressed	Tag
DISK		00:01:16	21-MAY-25	NO	TAG20250521T155031

List of Backup Pieces for backup set 3499 Copy #1

BP Key	Pc#	Status	Piece Name
3506	1	AVAILABLE	

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T155031_n2txxs03_bkp

3507	2	AVAILABLE	
------	---	-----------	--

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnndf_TAG20250521T155031_n2txztc3_bkp

BS Key	Size	Device	Type	Elapsed Time	Completion Time
3548	26.50K	DISK		00:00:00	21-MAY-25

BP Key: 3553 Status: AVAILABLE Compressed: NO Tag: TAG20250521T155205

Piece Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_anndf_TAG20250521T155205_n2ty0owd_bkp

List of Archived Logs in backup set 3548

Thrd	Seq	Low SCN	Low Time	Next SCN	Next Time
1	107	3206353	21-MAY-25	3206443	21-MAY-25

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
3591	Full		10.52M	DISK		00:00:01	21-MAY-25

BP Key: 3593 Status: AVAILABLE Compressed: NO Tag: TAG20250521T155207

Piece Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201708327_n2ty0qrz_bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3206472 Ckp time: 21-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

3680	Full	10.52M	DISK		00:00:00	21-MAY-25
------	------	--------	------	--	----------	-----------

BP Key: 3683 Status: AVAILABLE Compressed: NO Tag: TAG20250521T155532

Piece Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201708532_n2ty743s_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3206635 Ckp time: 21-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

3771	Full	10.52M	DISK		00:00:01	21-MAY-25
------	------	--------	------	--	----------	-----------

BP Key: 3774 Status: AVAILABLE Compressed: NO Tag: TAG20250521T155542

Piece Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201708542_n2ty7gwr_.bkp

SPFILE Included: Modification time: 20-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3206691 Ckp time: 21-MAY-25

RMAN> restore datafile 7;

Starting restore at 21-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=624 device type=DISK

channel ORA_DISK_1: restoring datafile 00007

input datafile copy RECID=10 STAMP=1201708528 file

name=**/home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2ty6tow_.dbf**

destination for restore of datafile 00007: /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00007, elapsed time: 00:00:15

output file name=/u01/app/oracle/oradata/ORCL/users01.dbf RECID=0 STAMP=0

Finished restore at 21-MAY-25

RMAN> recover datafile 7;

Starting recover at 21-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:03

Finished recover at 21-MAY-25

RMAN> alter database datafile 7 online;

Statement processed

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	900	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
3	690	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	TS1	NO	/u01/app/oracle/oradata/ORCL/ts01.dbf
7	153	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	120	TEMP	120	/u01/app/oracle/oradata/ORCL/temp01.dbf

- 장애 발생 - 모든 DFs

SQL> shut immediate

[oracle@oel7v9r1 ~]\$ rm /u01/app/oracle/oradata/ORCL/*.dbf

SQL> startup

ORACLE instance started.

Total System Global Area 1811936368 bytes

Fixed Size 8897648 bytes

Variable Size 436207616 bytes

Database Buffers 1358954496 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

RMAN> select status from v\$instance;

STATUS

MOUNTED

RMAN> restore database;

Starting restore at 21-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=745 device type=DISK

channel ORA_DISK_1: restoring datafile 00001

input datafile copy RECID=7 STAMP=1201708468 file

name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_system_n2ty43ty_.dbf

destination for restore of datafile 00001: /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00001, elapsed time: 00:00:46

output file name=/u01/app/oracle/oradata/ORCL/system01.dbf RECID=0 STAMP=0

channel ORA_DISK_1: restoring datafile 00003

input datafile copy RECID=8 STAMP=1201708495 file

name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_sysaux_n2ty56j0_.dbf

destination for restore of datafile 00003: /u01/app/oracle/oradata/ORCL/sysaux01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00003, elapsed time: 00:00:58

output file name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf RECID=0 STAMP=0

channel ORA_DISK_1: restoring datafile 00004

input datafile copy RECID=9 STAMP=1201708516 file

name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_undotbs1_n2ty5zxv_.dbf

destination for restore of datafile 00004: /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00004, elapsed time: 00:00:15

output file name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf RECID=0 STAMP=0

channel ORA_DISK_1: restoring datafile 00007

input datafile copy RECID=10 STAMP=1201708528 file

name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_users_n2ty6tow_.dbf

destination for restore of datafile 00007: /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00007, elapsed time: 00:00:03

```

output file name=/u01/app/oracle/oradata/ORCL/users01.dbf RECID=0 STAMP=0
channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/ts01.dbf
channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnnd0_TAG20250521T100158_n2t9j7g6_.bkp
channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_21/o1_mf_nnnd0_TAG20250521T100158_n2t9j7g6_.bkp
tag=TAG20250521T100158
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
Finished restore at 21-MAY-25

```

```

RMAN> recover database;

```

```

Starting recover at 21-MAY-25
using channel ORA_DISK_1
datafile 5 not processed because file is read-only

```

```

starting media recovery
media recovery complete, elapsed time: 00:00:01

```

```

Finished recover at 21-MAY-25

```

```

RMAN> alter database open;
Statement processed

```

```

SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	3409354	SYSTEM
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3409354	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3409354	ONLINE
5	TS1	/u01/app/oracle/oradata/ORCL/ts01.dbf		
3166776	ONLINE			
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf		
3409354	ONLINE			

```
SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	3409354	SYSTEM
2	DATA_TBS	/home/oracle/data_tbs01.dbf	3412250	ONLINE
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3409354	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3409354	ONLINE
5	TS1	/u01/app/oracle/oradata/ORCL/ts01.dbf	3166776	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	3409354	ONLINE

6 rows selected.

```
SQL> create table hr.march tablespace data_tbs as select * from hr.employees;
Table created.
```

```
SQL> select count(*) from hr.march;
COUNT(*)
```

```
-----
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```

- Image Copy로 특정 데이터파일의 백업을 위치 조정하며 진행

```
RMAN> backup as copy datafile 2 format '/u01/app/oracle/oradata/ORCL/data_tbs01.dbf';
```

```
Starting backup at 21-MAY-25
using channel ORA_DISK_1
channel ORA_DISK_1: starting datafile copy
input datafile file number=00002 name=/home/oracle/data_tbs01.dbf
output file name=/u01/app/oracle/oradata/ORCL/data_tbs01.dbf tag=TAG20250521T171050 RECID=13
STAMP=1201713050
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:01
Finished backup at 21-MAY-25
```

```
Starting Control File and SPFILE Autobackup at 21-MAY-25
piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201713052_n2v2ndc9_.bkp
comment=NONE
```

```
Finished Control File and SPFILE Autobackup at 21-MAY-25
```

```
SQL> !ls /u01/app/oracle/oradata/ORCL
```

```
bnr.sh control01.ctl data_tbs01.dbf redo02.log something.lst system01.dbf ts01.dbf users01.dbf
bnr1.sh control02.ctl redo01.log redo03.log sysaux01.dbf temp01.dbf undotbs01.울
```

RMAN> list copy of datafile 2;

List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
4010	2	A 21-MAY-25	3414512	21-MAY-25	NO

Name: /u01/app/oracle/oradata/ORCL/data_tbs01.dbf
Tag: TAG20250521T171050

- 장애 발생 - 특정 데이터파일(data_tbs01.dbf)

SQL> shut immediate

Database closed.

Database dismounted.

ORACLE instance shut down.

SQL> ! rm /home/oracle/data_tbs01.dbf

SQL> startup

ORACLE instance started.

Total System Global Area 1811936368 bytes

Fixed Size 8897648 bytes

Variable Size 436207616 bytes

Database Buffers 1358954496 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 2 - see DBWR trace file

ORA-01110: data file 2: '/home/oracle/data_tbs01.dbf'

SQL> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR
CHANGE#	TIME	CON_ID
2	ONLINE	ONLINE FILE NOT FOUND
0		0

SQL> alter database datafile 4 offline;

Database altered.

RMAN> list copy of datafile 2;

List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
4010	2	A 21-MAY-25	3414512	21-MAY-25	NO

Name: /u01/app/oracle/oradata/ORCL/data_tbs01.dbf
Tag: TAG20250521T171050

RMAN> switch datafile 2 to copy;

datafile 2 switched to datafile copy "/u01/app/oracle/oradata/ORCL/data_tbs01.dbf"
starting full resync of recovery catalog
full resync complete

RMAN> recover datafile 2;

Starting recover at 21-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=744 device type=DISK

starting media recovery
media recovery complete, elapsed time: 00:00:00

Finished recover at 21-MAY-25

RMAN> alter database open;

Statement processed

RMAN> alter database datafile 2 online;

Statement processed

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	900	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	5	DATA_TBS	NO	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
3	690	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	TS1	NO	/u01/app/oracle/oradata/ORCL/ts01.dbf
7	153	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	120	TEMP	120	/u01/app/oracle/oradata/ORCL/temp01.dbf

- user managed로 백업 받은 파일

SQL> create tablespace insa_tbs datafile '/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf' size 5m;
 Tablespace created.

SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a,
 v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	3415103	SYSTEM
2	DATA_TBS	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf	3415103	ONLINE
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3415103	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3415100	ONLINE
5	TS1	/u01/app/oracle/oradata/ORCL/ts01.dbf	3166776	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	3415103	ONLINE
8	INSA_TBS	/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf	3416088	ONLINE

7 rows selected.

SQL> create table hr.emp3 tablespace insa_tbs as select * from hr.employees;
 Table created.

[oracle@oel7v9r1 datafile]\$ cp -v /u01/app/oracle/oradata/ORCL/insa_tbs01.dbf

```
/home/oracle/backup/FRA/ORCL/datafile/
```

```
'/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf' -> '/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf'
```

```
SQL> !ls /home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf
```

```
/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.을
```

- 수동으로 백업받은 파일을 rman image copy로 등록

```
RMAN> report need backup;
```

```
starting full resync of recovery catalog
```

```
full resync complete
```

```
RMAN retention policy will be applied to the command
```

```
RMAN retention policy is set to redundancy 1
```

```
Report of files with less than 1 redundant backups
```

```
File #bkps Name
```

```
-----
```

2	0	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
---	---	---

8	0	/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf
---	---	---

```
RMAN> catalog datafilecopy '/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf';
```

```
cataloged datafile copy
```

```
datafile copy file name=/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf RECID=14 STAMP=1201717591
```

```
RMAN> report need backup;
```

```
RMAN retention policy will be applied to the command
```

```
RMAN retention policy is set to redundancy 1
```

```
Report of files with less than 1 redundant backups
```

```
File #bkps Name
```

```
-----
```

2	0	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
---	---	---

```
RMAN> backup as copy datafile 2;
```

```
Starting backup at 21-MAY-25
```

```
using channel ORA_DISK_1
```

```
channel ORA_DISK_1: starting datafile copy
```

```
input datafile file number=00002 name=/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
```

```
output file name=/home/oracle/backup/FRA/ORCL/datafile/o1_mf_data_tbs_n2v73lf2_.dbf
```

tag=TAG20250521T182714 RECID=15 STAMP=1201717637
channel ORA_DISK_1: datafile copy complete, elapsed time: 00:00:07
Finished backup at 21-MAY-25

Starting Control File and SPFILE Autobackup at 21-MAY-25
piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201717641_n2v73t18_.bkp
comment=NONE
Finished Control File and SPFILE Autobackup at 21-MAY-25

RMAN> change datafilecopy '/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf' uncatalog;
uncataloged datafile copy
datafile copy file name=/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf RECID=14 STAMP=1201717591
Uncataloged 1 objects

RMAN> list copy of tablespace insa_tbs;
specification does not match any datafile copy in the repository

RMAN> catalog datafilecopy '/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf';
cataloged datafile copy
datafile copy file name=/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf RECID=16 STAMP=1201717791

RMAN> list copy of tablespace insa_tbs;
List of Datafile Copies

=====

Key	File S	Completion Time	Ckp SCN	Ckp Time	Sparse
4200	8	A 21-MAY-25	3422331	21-MAY-25	NO
Name: /home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf					

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File Size(MB)	Tablespace	RB segs	Datafile Name
1	900	SYSTEM	YES /u01/app/oracle/oradata/ORCL/system01.dbf

2	5	DATA_TBS	NO	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
3	690	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	TS1	NO	/u01/app/oracle/oradata/ORCL/ts01.dbf
7	153	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	5	INSA_TBS	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	120	TEMP	120	/u01/app/oracle/oradata/ORCL/temp01.dbf

-장애 발생

```
SQL> !rm /u01/app/oracle/oradata/ORCL/insa_tbs01.dbf
```

```
SQL> !ls /home/oracle/backup/FRA/ORCL/datafile
```

```
insa_tbs01.dbf          o1_mf_sysaux_n2ty56j0_.dbf  o1_mf_undotbs1_n2ty5zxv_.dbf
o1_mf_data_tbs_n2v73lf2_.dbf  o1_mf_system_n2ty43ty_.dbf  o1_mf_users_n2ty6tow_.dbf
```

```
SQL> !ls /u01/app/oracle/oradata/ORCL/
```

```
bnr.sh  control01.ctl  data_tbs01.dbf  redo02.log  something.lst  system01.dbf  ts01.dbf  users01.dbf
bnr1.sh  control02.ctl  redo01.log      redo03.log  sysaux01.dbf  temp01.dbf  undotbs01.울
```

```
RMAN> alter tablespace insa_tbs offline immediate;
```

Statement processed

```
RMAN> report schema;
```

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	900	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	5	DATA_TBS	NO	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
3	690	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf

5	10	TS1	NO	/u01/app/oracle/oradata/ORCL/ts01.dbf
7	153	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	5	INSA_TBS	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	120	TEMP	120	/u01/app/oracle/oradata/ORCL/temp01.dbf

RMAN> restore tablespace insa_tbs;

Starting restore at 21-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: restoring datafile 00008

input datafile copy RECID=16 STAMP=1201717791 file

name=/home/oracle/backup/FRA/ORCL/datafile/insa_tbs01.dbf

destination for restore of datafile 00008: /u01/app/oracle/oradata/ORCL/insa_tbs01.dbf

channel ORA_DISK_1: copied datafile copy of datafile 00008, elapsed time: 00:00:04

output file name=/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf RECID=0 STAMP=0

Finished restore at 21-MAY-25

RMAN> recover tablespace insa_tbs;

Starting recover at 21-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:02

Finished recover at 21-MAY-25

RMAN> alter tablespace insa_tbs online;

Statement processed

RMAN> report schema;

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name

1	900	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	5	DATA_TBS	NO	/u01/app/oracle/oradata/ORCL/data_tbs01.dbf
3	690	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	TS1	NO	/u01/app/oracle/oradata/ORCL/ts01.dbf
7	153	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	5	INSA_TBS	NO	/u01/app/oracle/oradata/ORCL/insa_tbs01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name

1	120	TEMP	120	/u01/app/oracle/oradata/ORCL/temp01.dbf

```

RMAN> report obsolete;
RMAN> delete obsolete;
RMAN> crosscheck copy;
RMAN> list expired copy;
RMAN> delete expired copy;
RMAN> crosscheck archivelog all;
RMAN> list expired archivelog all;
RMAN> delete expired archivelog all;

```

<시나리오 20> 파라미터 파일(spfile, pfile)이 손상되었을 경우 복구

```
SQL> select dbid, name from v$database;
```

```
DBID NAME
```

```
-----
```

```
1727762462 ORCL
```

```
RMAN> backup database plus archivelog;
```

-장애 발생

```
[oracle@oel7v9r1 ~]$ rm -rf $ORACLE_HOME/dbs/initorcl.ora
```

```
[oracle@oel7v9r1 ~]$ rm -rf $ORACLE_HOME/dbs/spfileorcl.ora
```

-재접속

```
[oracle@oel7v9r2 ~]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

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connected to target database (not started)

connected to recovery catalog database

```
RMAN> startup force nomount
```

← 강제 nomount

startup failed: ORA-01078: failure in processing system parameters

LRM-00109: could not open parameter file '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/initorcl.ora'

starting Oracle instance without parameter file for retrieval of spfile

Oracle instance started

Total System Global Area 1073737800 bytes

Fixed Size 8904776 bytes

Variable Size 289406976 bytes

Database Buffers 767557632 bytes

Redo Buffers 7868416 bytes

```
RMAN> set dbid 1727762462
```

executing command: SET DBID

database name is "ORCL" and DBID is 1727762462

RMAN> restore spfile from autobackup;

Starting restore at 21-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250521

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250520

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250519

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250518

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250517

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250516

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20250515

channel ORA_DISK_1: no AUTOBACKUP in 7 days found

RMAN-00571: =====

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====

RMAN-00571: =====

RMAN-03002: failure of restore command at 05/21/2025 19:11:09

RMAN-06172: no AUTOBACKUP found or specified handle is not a valid copy or piece <- 잘 못찾는다.

RMAN> restore spfile from

'/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201719478_n2v8x61p_.bkp';

➔ 그래서 지정

Starting restore at 21-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: restoring spfile from AUTOBACKUP

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_21/o1_mf_s_1201719478_n2v8x61p_.bkp

channel ORA_DISK_1: SPFILE restore from AUTOBACKUP complete

Finished restore at 21-MAY-25

RMAN> shutdown abort

Oracle instance shut down

RMAN> startup

connected to target database (not started)

Oracle instance started

database mounted

database opened

Total System Global Area 1811936368 bytes

Fixed Size	8897648 bytes
Variable Size	436207616 bytes
Database Buffers	1358954496 bytes
Redo Buffers	7876608 bytes

<시나리오 21> Incremental backup